

Ashish Karhale

Software Developer

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Technical Skills

AI Infrastructure: vLLM, Llama.cpp, RAG, LangGraph, OpenAI

LLM Training: Fine-tuning (LoRA/QLoRA), Prompt Engineering

Backend & Middleware: FastAPI, Django, Redis, REST APIs

Cloud Services: AWS EC2, AWS RDS, S3, Docker, CI/CD

Data Processing: Pandas, NumPy, Spark, Matplotlib

Databases: PostgreSQL, pgvector, SQLAlchemy, Redis

Distributed Systems: WebSocket, RAG Pipelines, Celery

Concepts: OOP, JWT, System Optimization, Design Patterns

Education

National Institute of Technology (NIT), Surat

Surat, Gujarat

Bachelor of Technology (B.Tech)

May 2021

Work Experience

Software Developer

Zeron

Oct 2025 – Present

- Developed a GRC platform that reduced control mapping processing time from **7 days to 10 minutes** through automated mapping engines.
- Built high-throughput AI inference stacks using **vLLM** and **Llama.cpp** deployed on **EC2** instances for local and cloud-based LLM serving.
- Trained and fine-tuned specialized models for cybersecurity frameworks using **LoRA/QLoRA** techniques and **OpenAI**.
- Implemented low-latency caching and message brokering using **Redis** to manage real-time platform telemetry and entity mentions.
- Engineered advanced RAG pipelines with **LangGraph** and **LangChain**, utilizing **pgvector** on **RDS** for native embeddings in PostgreSQL.
- Designed and maintained **REST APIs** to power an extensible commenting system and customized security frameworks.
- Tech Stack:** Python, FastAPI, PostgreSQL, RDS, Redis, vLLM, Llama.cpp, LangGraph, LangChain, EC2, Pandas.

Full Stack Developer

Cyethack Security Solution Pvt. Ltd.

Oct 2023 – Oct 2025

- Architected and developed a scalable remote auditing system managing **100K+ nodes** with real-time monitoring.
- Optimized system throughput, achieving **40x improvement** (30K to 1200K audits per day) via reliable workflows.
- Reduced CPU usage from **90% to 34%** with the help of proper **data structures** and efficient **algorithms**.
- Developed and secured **REST APIs** for a comprehensive CIA security framework with encrypted data transfer.
- Built interactive analytics dashboard with role-based access control, providing insights into node vulnerabilities.
- Tech Stack:** Python, Django, React, Docker, Django Rest Framework, REST APIs, SQL, JavaScript, WebSocket.

Technical Projects

IPL Data Analysis Application

GitHub

- Developed a comprehensive data analytics application analyzing 12 seasons of IPL data with 225K+ records.
- Implemented advanced data preprocessing and statistical analysis to extract insights about player performance.
- Created interactive dashboards for player comparison, team analysis, and player strengths/weaknesses.
- Tech Stack:** Python, Streamlit, Pandas, Plotly, Seaborn, Matplotlib, Jupyter Notebook, Linux.

Real-time Chat Application

GitHub

- Built a feature-rich real-time chat application supporting private and group conversations with instant delivery.
- Implemented secure JWT-based authentication system with user profile management and session handling.
- Developed robust **REST APIs** via **Django Rest Framework** and integrated **WebSocket** connections via **Django Channels**.
- Tech Stack:** Python, Django, React, REST APIs, Django Channels, JWT, WebSocket, HTML/CSS.

Achievements

- System Optimization:** 40x throughput improvement in production via algorithmic and concurrency optimization.
- Competitive Programming:** 250+ problems solved on LeetCode and GeeksforGeeks.
- Leadership:** Successfully led and mentored teams in high-pressure technical environments.